Project Report

Class: FA23-BCS (Section B)

Course: Object Oriented Programming

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Project Title

Bank Management System

Project Overview: The Bank Management System is designed to handle various user roles, including Manager, CSR (Customer Service Representative), and Account Holder (formerly User). The system uses a graphical user interface (GUI) built with Java Swing, where different dashboards provide access to relevant features depending on the user type.

Functional Requirements

1. Manager Dashboard

- The Manager can log in using a unique Manager ID and password.
- The Manager can:
  - Add a new CSR.
  - Delete an existing CSR.
  - View a list of all Account Holders (view user details).

2. CSR Dashboard

- The CSR logs in using their CSR ID and password.
- The CSR can:
  - Create a new account holder (Account Holder).
  - Delete an existing account holder.
  - View account holder details.
  - Deposit money into an account holder's account.

3. Account Holder Dashboard

- The Account Holder logs in using their Account Holder ID and password.
- The Account Holder can:
  - Create an account (only if a new account holder).

- Withdraw money from their account.
- View transaction history.
- View their account balance.

User Interface Design

1. Manager Login Frame

- **Elements:** Labels for Manager ID and Password, text fields for entering credentials, Sign-In button.
- **Actions:** On correct login, Manager Dashboard is displayed. On incorrect login, an error message is shown.

2. CSR Login Frame

- **Elements:** Labels for CSR ID and Password, text fields for entering credentials, Sign-In button.
- **Actions:** On correct login, CSR Dashboard is displayed. On incorrect login, an error message is shown.

3. Account Holder Login Frame

- **Elements:** Labels for Account Holder ID and Password, text fields for entering credentials, Sign-In button.
- **Actions:** On correct login, Account Holder Dashboard is displayed. On incorrect login, an error message is shown.

4. Manager Dashboard

- **Elements:** Label displaying "Dashboard", buttons for adding and deleting CSRs, viewing account holders.
- **Actions:** Clicking "Add CSR" opens the AddCSR frame. Clicking "Delete CSR" opens the DeleteCSR frame. Clicking "View Account Holders" shows a list of all Account Holders.

5. CSR Dashboard

- **Elements:** Label displaying "Dashboard", buttons for creating accounts, deleting accounts, viewing account holders, and depositing money.
- **Actions:** Clicking buttons triggers respective actions like creating, deleting accounts, viewing account holder details, and depositing money.

6. Account Holder Dashboard

- **Elements:** Label displaying "Account Holder Dashboard", buttons for creating an account, withdrawing money, viewing transaction history, and viewing account balance.

- **Actions:** Clicking buttons opens frames for creating an account, withdrawing money, viewing transactions, and checking account balance.

Implementation

1. Manager Dashboard

- The ManagerDashBoardDisplay() method initializes the Manager Dashboard with buttons for adding and deleting CSRs, and viewing account holders. Action listeners trigger respective frames.

2. CSR Dashboard

- The CSRDashBoardDisplay() method initializes the CSR Dashboard with buttons for creating, deleting accounts, viewing account holders, and depositing money. Action listeners trigger methods such as createAccount(), deleteAccount(), viewAccount(), and depositAmount().

3. Account Holder Dashboard

- The AccountHolderDashBoardDisplay() method initializes the Account Holder Dashboard with buttons for creating an account, withdrawing money, viewing transaction history, and viewing account balance. Action listeners trigger methods like createAccount(), withdrawMoney(), viewTransactionHistory(), and viewBalance().

4. Login Frames

- The login methods (managerLoginFrame(), CSRLoginFrame(), accountHolderLoginFrame()) handle credential verification and redirect the user to the appropriate dashboard upon successful login.

5. CSR Operations

- createAccount(): Creates a new account holder.

- deleteAccount(): Deletes an existing account holder.

- viewAccount(): Displays account holder details.

- depositAmount(): Deposits money into an account holder’s account.

6. Add and Delete CSR

- csrAddFrame() and csrDeleteFrame() methods open frames for adding and deleting CSRs.

Key Functionalities

1. Manager

- **Login:** The Manager logs in with a unique ID and password.
- **Manage CSRs:** The Manager can add or delete CSRs from the dashboard.
- **View Account Holders**: The Manager can view a list of all Account Holders.

2. CSR

- **Login:** The CSR logs in with their ID and password**.**
- **Manage Account Holders:** The CSR can create, delete, view, and deposit money into account holders' accounts.

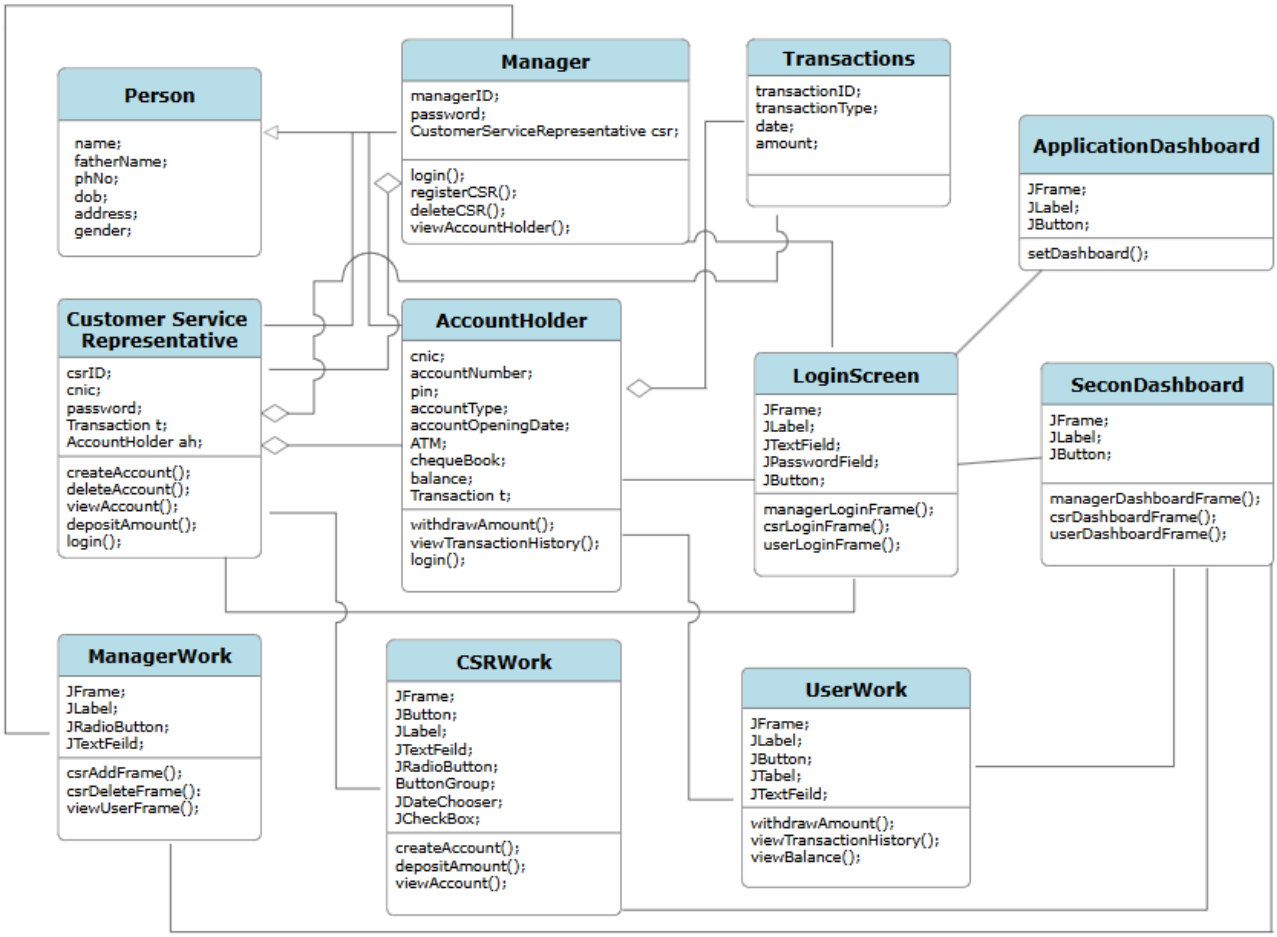
3. Account Holder

- **Login:** The Account Holder logs in with their ID and password.
- **Account Management:** The Account Holder can create an account, withdraw money, view transaction history, and view their account balance.

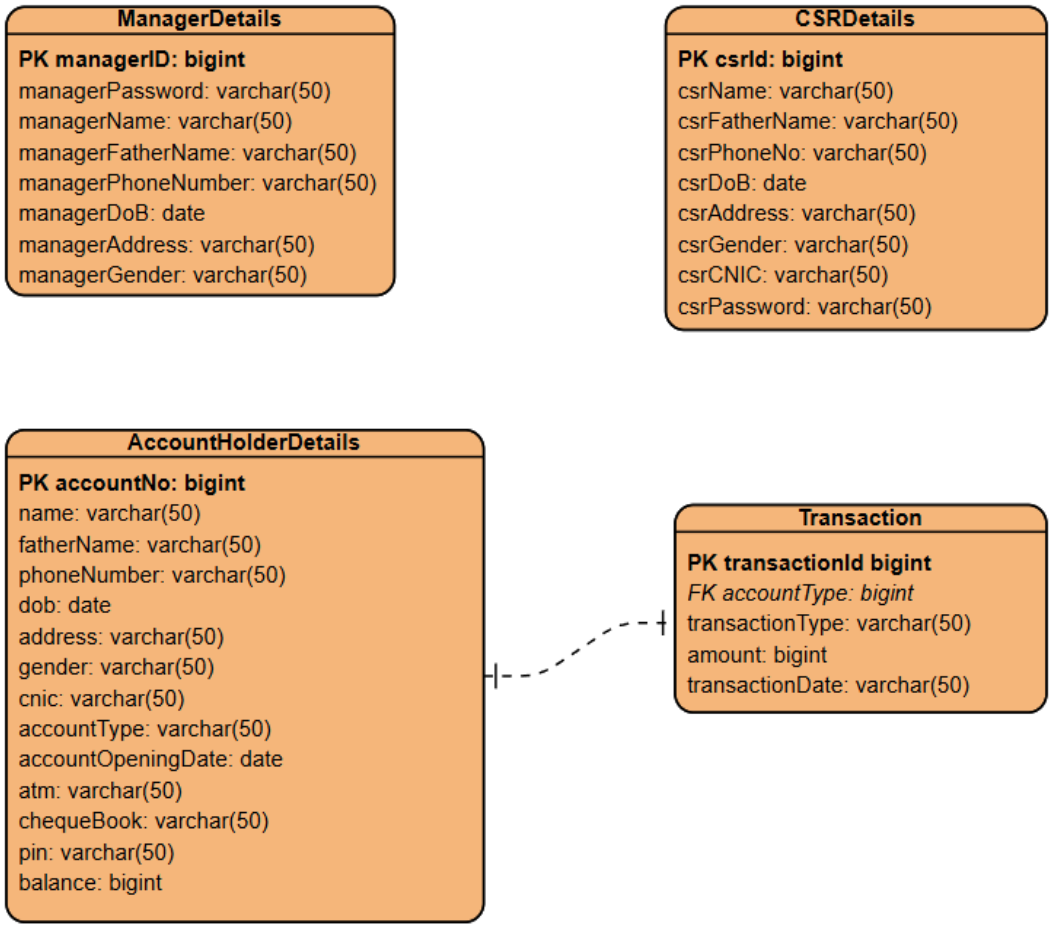
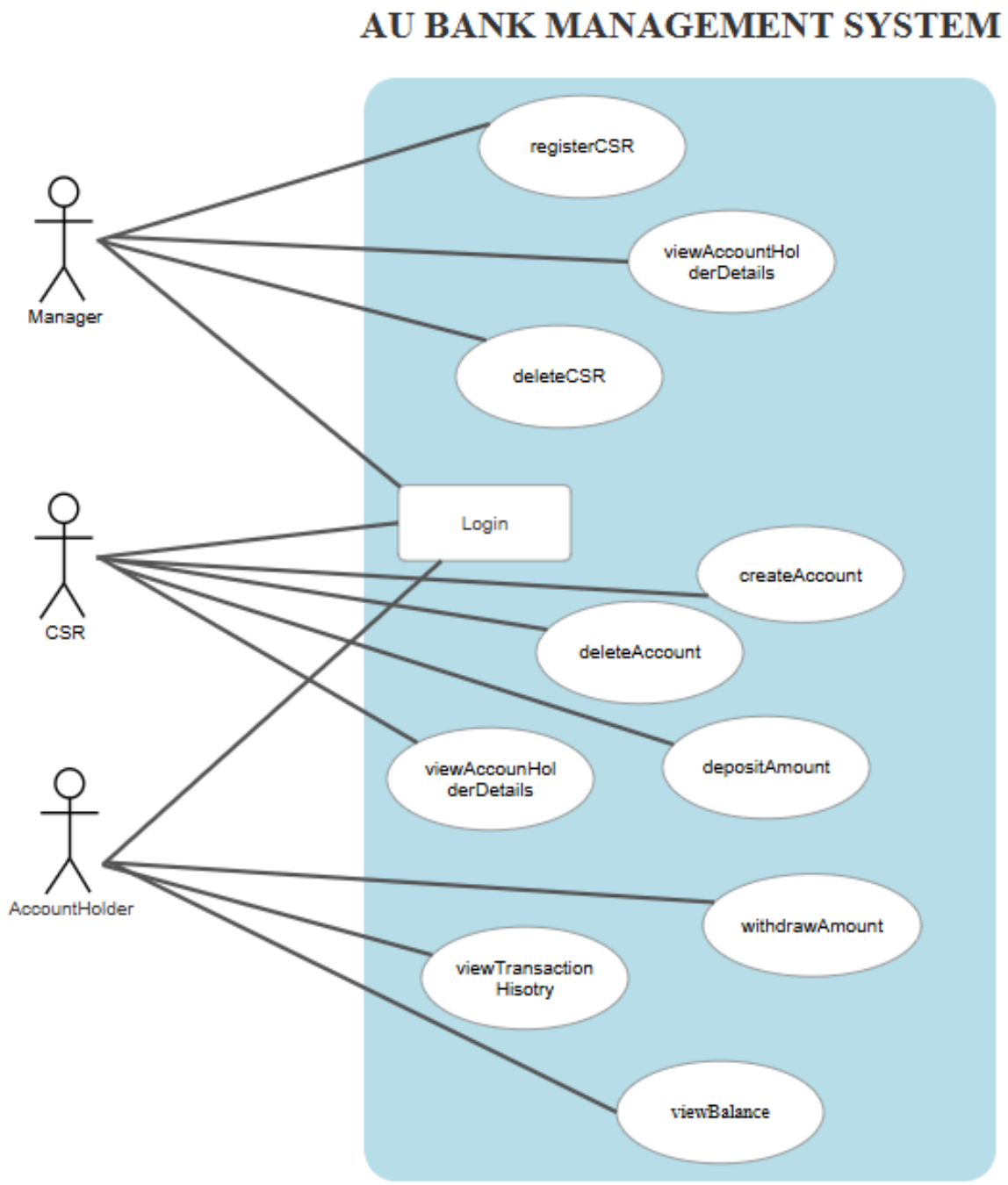
Error Handling

- **Login Errors:** Invalid credentials trigger error messages (e.g., "Username or password is incorrect").
- **Input Validation:** The system ensures valid IDs are entered for Manager and CSR logins, and prompts users if invalid input is detected.

UML (Unified Modeling Language)



ERD (Entity Relationship Diagram)



USE CASE DIAGRAM